CS5310_assignment8

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Abstract

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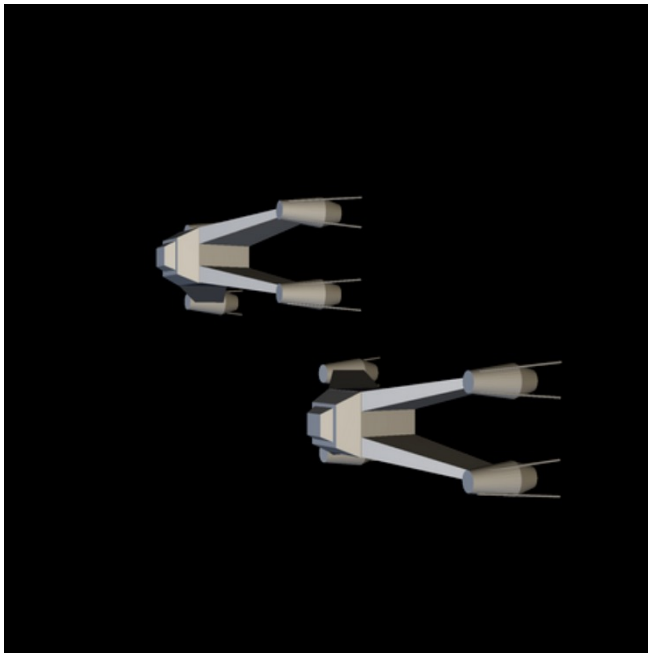

Figure 1: Enter Caption

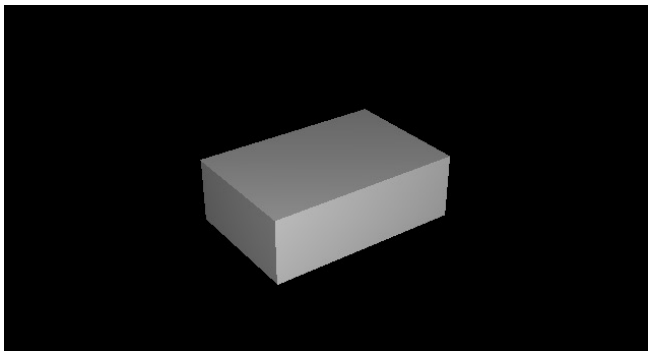

Figure 2: Enter Caption

1 Chapter 1

All the basic

2 Chapter 2

$h(x, z, t) = -y_0 + \sum_{i=1}^{N_w} A_i \cos(k_{ix}x + k_{iz}z - \omega_i t)$

This week, I successfully implemented Sinusoids Wave and manually constructed a mesh, along with a real-time rendering model. This allows me to render the ocean surface in real time. Currently, since it operates on a single thread, the efficiency is quite low. I can only manage to render 2500 triangular surfaces to achieve real-time rendering. However, I have verified the results using multiple methods and confirmed their accuracy. I have also uploaded a GIF to demonstrate the results, as it couldn't be included in the PDF due to format limitations.

Furthermore, during the implementation, I faced several challenges, especially in optimizing the performance for real-time rendering. By manually constructing the mesh, I ensured greater control over the rendering process, which helped in debugging and fine-tuning the wave simulation. The single-threaded approach, while limiting in performance, allowed for a more straightforward validation of the correctness of the wave patterns and their interactions.

The GIF I uploaded visually illustrates the wave dynamics and how the sinusoidal functions are effectively simulating the ocean surface. Although the current setup restricts the number of triangles, the foundational work is solid and can be scaled with multi-threading or GPU acceleration in future iterations to handle more complex and detailed surfaces.

Figure 3

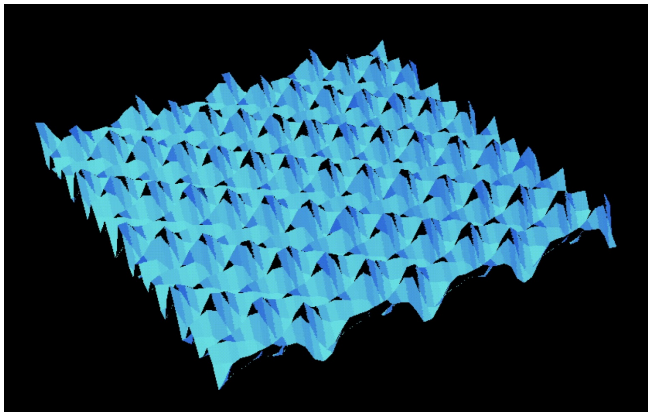

Figure 4: Enter Caption

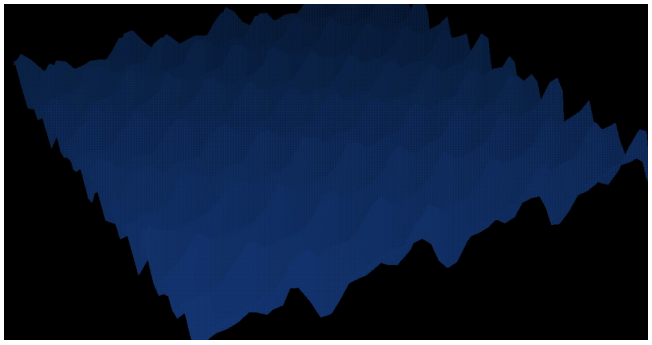

Figure 5: Enter Caption

3 what I do and leanrt

I have completed the implementation of the basic lighting techniques and attempted to automatically generate triangular models, along with automatically generating normals. Based on our oneside approach, I calculated the normals' order through the points of the triangles. Additionally, I have simulated the basic implementation of water flow. Moving forward, I will be focusing on implementing more advanced water simulations, as well as incorporating advanced techniques such as textures.

In detail, the basic lighting techniques I implemented include ambient, diffuse, and specular lighting. These foundational techniques are crucial for rendering realistic surfaces and have been integrated seamlessly into the rendering pipeline. The automatic generation of triangular models has significantly streamlined the modeling process, allowing for more efficient and consistent mesh creation. By calculating normals automatically, I have ensured that the lighting calculations are accurate and that the models respond correctly to various light sources.

The basic water flow simulation was a challenging yet rewarding process. It involved creating dynamic interactions between the generated triangles to mimic the natural movement of water. This initial implementation sets the stage for more sophisticated simulations, which will include more complex behaviors and interactions. Advanced water simulation techniques will help achieve more realistic and visually appealing results.

Furthermore, I plan to enhance the visual quality of the simulations by incorporating textures. This will involve mapping texture images onto the triangular surfaces, which will add depth and realism to the rendered scenes. Using textures will also allow for more detailed and varied visual effects, contributing to a richer and more immersive simulation.

In conclusion, these developments mark significant progress in the project. The combination of basic lighting techniques, automatic model and normal generation, and the initial water flow simulation provides a strong foundation. The upcoming implementation of advanced water simulations and textures will further elevate the quality and realism of the project, making it more sophisticated and visually compelling.